Colorimetric Analysis Methodology

Bacterial-Dye-Antibiotic Arrays

March 13, 2026

Problem Setup

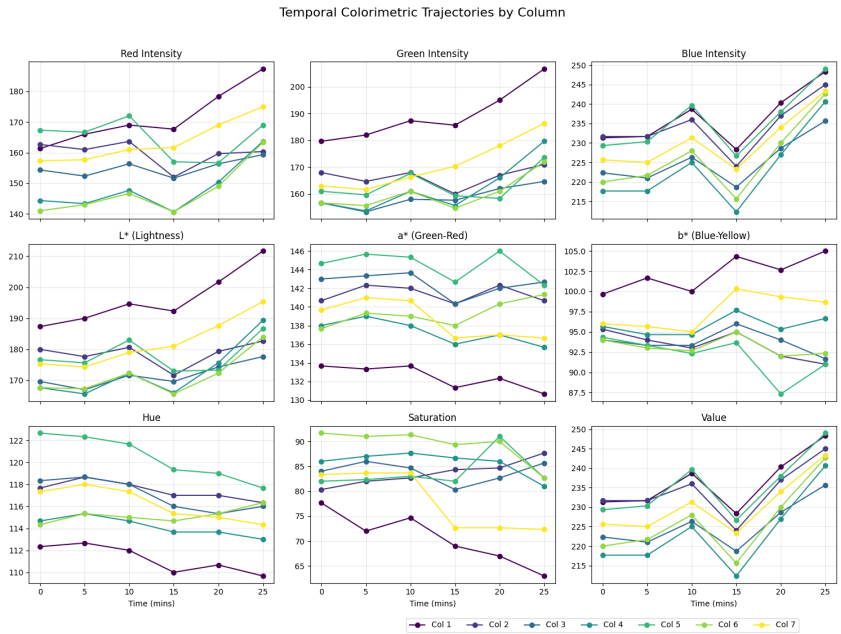
- **Objective**: Extract biological insights from a temporal sequence of colorimetric images (0th min to 25th min).
- **System**: Bacterial-dye-antibiotic arrays (3x9 grid matrix).
- **Goal**: Determine the Minimum Inhibitory Concentration (MIC) and understand reaction kinetics.
- **Indicator**: Biological indicator dye transitions from blue (high a^*) to pink (higher a^*) as surviving bacteria metabolize it.

Grid Detection & Extraction

- **Histogram Projection**: Isolate bright sample wells from the dark tray background using grayscale conversion, thresholding, and morphology opening.
- **Peak Finding**: Identify well centers via prominent peaks in projected 1D histograms over X and Y axes.
- **Median Extraction**: Draw a 32-pixel radius circular mask at each identified well location and extract median intensity across RGB, CIELAB ($L^*a^*b^*$), and HSV color spaces.

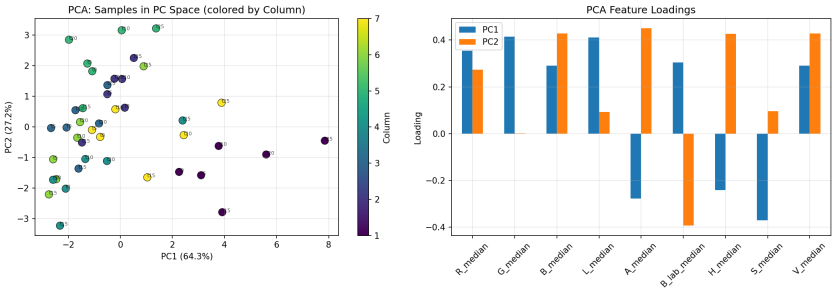
1. Temporal MIC Identification

- **Method**: Automated kinetic slope detection on the a^* (green-red) channel over time.
- **Threshold**: Monotonic linear regression slope ($m > 0.1$) indicates bacterial survival.
- **Result**: Column 5 identified as the MIC (last completely inhibited concentration).



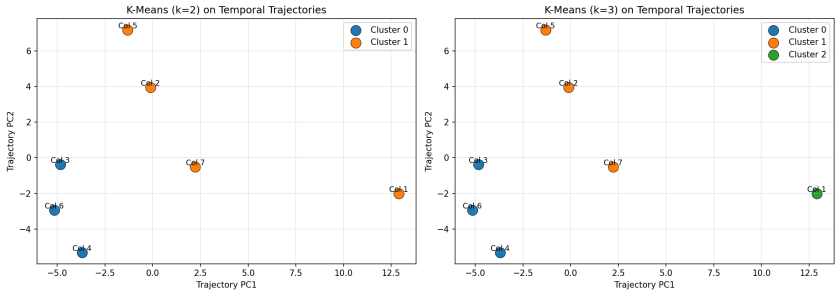
2. Principal Component Analysis (PCA)

- **Purpose**: Dimensionality reduction of 9 distinct color features per sample per timepoint.
- **Inference**: Green (G), Lightness (L^*), and Saturation (S) channels carry the highest variance over the temporal sequence, more than traditional red-blue axes.



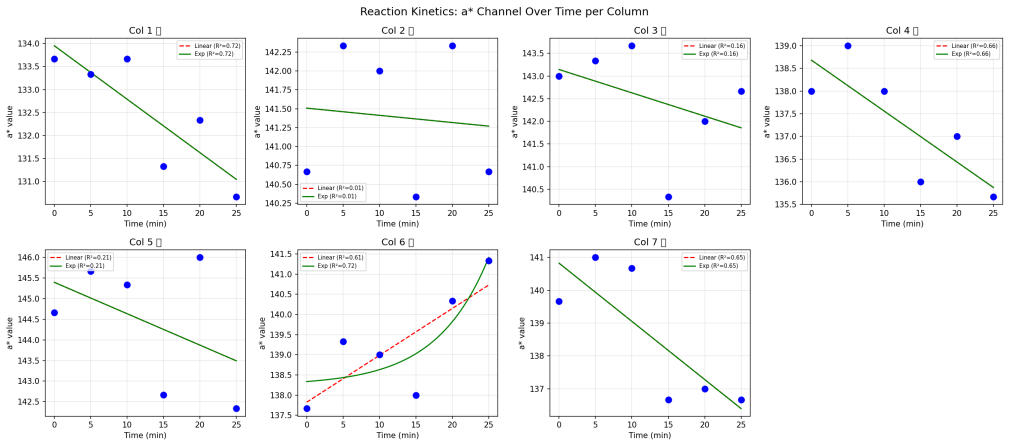
3. K-Means Behavioral Clustering

- **Method**: Clustering flattened temporal trajectories ($K = 3$).
- **Clusters identified**:
  1. High Inhibition (Col 1)
  2. Transition/Stable Zone (Cols 2, 5, 7)
  3. High Activity (Cols 3, 4, 6)
- **Inference**: Response is not a smooth gradient but forms discrete "plateaus" of efficacy.



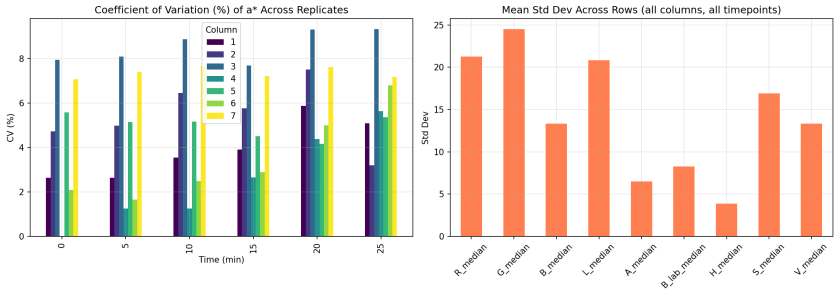
4. Early Prediction & Curvilinear Kinetics

- **Early Forecasting**: Logistic regression classifier achieved 85.7% accuracy predicting final MIC outcome at $t = 25$ using *only* the initial $t = 0$ image.
- **Curvilinear Kinetics**: Extracting R^2 exponential decay/growth coefficients using non-linear curve fits to overlying a^* channel datapoints. Provides insight into deceleration rate and specific bacteriostatic/bactericidal mechanism.



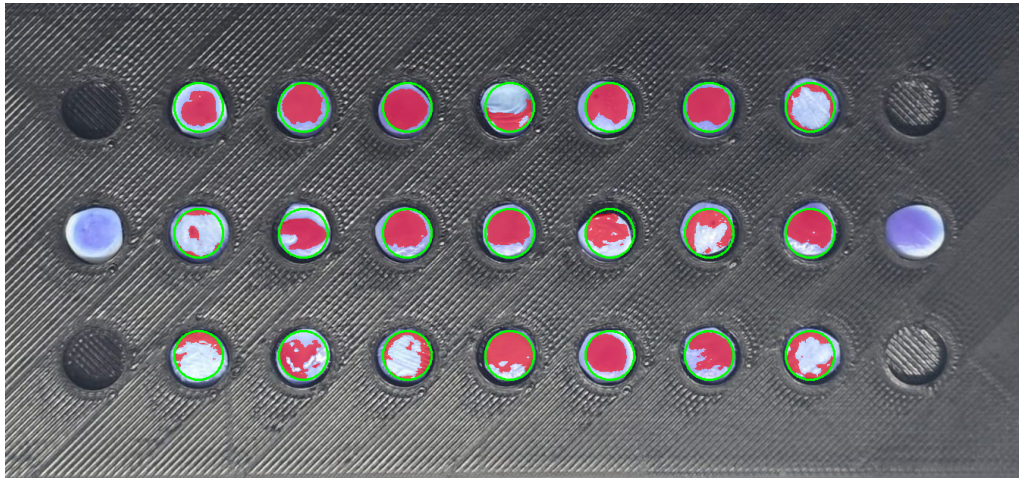
5. Replicate Consistency

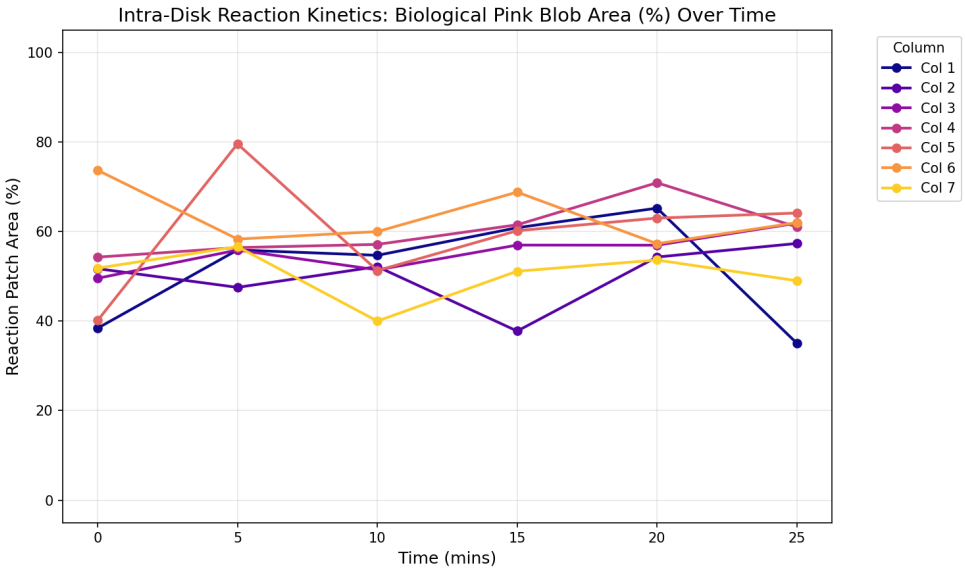
- **Method**: Calculated Coefficient of Variation (CV%) across the 3 replicate rows.
- **Inference**: Assay layout is highly stable (CV 2.5% - 5.5%).
- **Anomaly**: Column 3 exhibited a variance spike (8.54% CV), a hallmark of borderline transition to near-lethal antibiotic stresses.



6. Intra-Disk Region Detection

- **Method**: K-Means clustering ($K = 2$) spatially *within* pixels of sample wells to isolate 'Reacted' (pink) from 'Unreacted' (blue).
- **Inference**: Growth isn't a uniform color fade. Pink region area percentage expands non-linearly, showing localized metabolic activity.





Conclusion

- Successfully automated MIC identification using robust, multi-channel colorimetric analysis.
- Advanced ML (PCA, K-Means) revealed deeper color channel insights & non-linear behavioral responses.
- Spatial segmentation within discs proves growth is localized before diffusing.
- Early prediction models suggest feasibility of significantly reducing required incubation time.